

Software Requirement Specification (SRS)

Project Title

Mathematical Customer Lifetime Value (CLV) Projection using BG/NBD Model from Scratch

1. Introduction

This project aims to predict **Customer Lifetime Value (CLV)** using the **BG/NBD (Beta-Geometric/Negative Binomial Distribution)** model. The model is implemented from scratch without using pre-built marketing libraries. It analyzes customer transaction history to estimate future purchases and customer value.

2. Objectives

- Implement the BG/NBD mathematical model from scratch.
 - Estimate the parameters (r , α , a , b) using the L-BFGS-B optimization algorithm.
 - Predict future customer purchases.
 - Calculate Customer Lifetime Value (CLV).
 - Generate reports and visualizations for analysis.
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3. Functional Requirements

- Upload customer transaction dataset (CSV).
 - Preprocess and clean the data.
 - Compute the BG/NBD likelihood function.
 - Estimate model parameters using SciPy L-BFGS-B.
 - Predict future transactions.
 - Calculate Customer Lifetime Value.
 - Display graphs and export results.
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4. Non-Functional Requirements

- Fast processing for large datasets.
 - Accurate mathematical predictions.
 - Easy-to-use interface.
 - Secure handling of customer data.
 - Modular and maintainable code.
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5. Software & Hardware Requirements

Software

- Python 3.11+
- Jupyter Notebook / VS Code
- NumPy
- Pandas
- SciPy
- Matplotlib
- Seaborn

Hardware

- Intel Core i3 or above
 - 8 GB RAM
 - 20 GB Free Storage
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6. Input and Output

Input

- Customer ID
- Transaction Date
- Purchase Frequency
- Recency
- Observation Period (T)

Output

- Estimated BG/NBD Parameters (r , α , a , b)
- Future Purchase Prediction
- Customer Lifetime Value (CLV)
- Charts and Summary Report

7. Advantages

- Predicts future customer purchasing behavior.
 - Helps improve customer retention.
 - Supports business decision-making.
 - Works efficiently with large transaction datasets.
 - Uses mathematical optimization for better accuracy.
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8. Conclusion

The **Mathematical Customer Lifetime Value Projection using the BG/NBD Model from Scratch** is an effective analytical system for predicting customer purchasing behavior. By estimating model parameters through mathematical optimization, it provides accurate CLV predictions that help businesses improve marketing strategies and customer relationship management.